

Campaign Level

Budget pacing determines spending pressure

Campaign state

- Remaining budget
- Current pacing reference
- Time and delivery status

Campaign control

(e.g., traffic-aware, PID, dual)

a_{it}^{ref}

Opportunity Level

Autobidder evaluates the current impression

Impression features

- User and context
- Ad and creative features
- Expected value / conversion

Autobidder

(e.g., learned bidder)

a_{it}^{raw}

Execution boundary κ

$$a_{it}^{exec} = clip(a_{it}^{raw}, \max\{0, (1 - \kappa)a_{it}^{ref}\}, (1 + \kappa)a_{it}^{ref})$$

Auction Environment

Many advertisers compete for each impression

Advertiser 1

Advertiser 2

... Advertiser N

Real-time auction

(e.g., second price)

a_{it}^{exec}

Winner

1. Value-Delivery Frontier

How does execution discretion affect campaign outcomes?

Campaign
(state, goals, targeting)

Execution
width κ

Value
(e.g., conversions, CPA, ROAS)

Delivery
(e.g., spend rate, budget delivery)

2. Drivers of Discretion Value

When does wider discretion help?

Incremental selection information

Campaign load

Selection room

3. Platform Deployment and Competitive Spillovers

What happens when many advertisers adopt execution discretion?

Advertiser 1

Advertiser 2

... Advertiser N

Shared marketplace
(real-time auction)

Ad allocation

Clearing prices

Platform revenue

Spillovers